HW Portfolio

1. PIM · AXI SoC 통합 설계

평균 연산 사이클 7,673 → 316

2. Ising Machine (FPGA)

사이클 11.4배 단축 · 타이밍 마진 7.67% · UVM 620회 회귀

3. Flash ADC

ENOB 2.95bit · INL 0.4mV

4. 8-bit SAR ADC

20MHz · ENOB 7.75bit

5. MIPS CPU

CLA 적용으로 연산 지연 76%↓ · 타이밍 마진 33%↑

최세현

PROJECT 01. 공정 미세화 한계 완화를 위한 PIM 개념의 Near-Memory 하드웨어 가속기 및 AXI 기반 SoC 통합 설계

- 수행기간

  2025.12~2026.06

- 수행인원

  3명 (스펙설정 및 모델링, RTL코딩 인터페이스 통합 및 FPGA 디버깅, Testbench 검증 및 보드 테스트)

- 사용기술

  VIVADO(verilog), FPGA, ILA

- 수행내용

  (1) Bottom to top으로 16bit CPU설계 및 UART로 들어오는 입력을 처리하는 제어 로직 구현

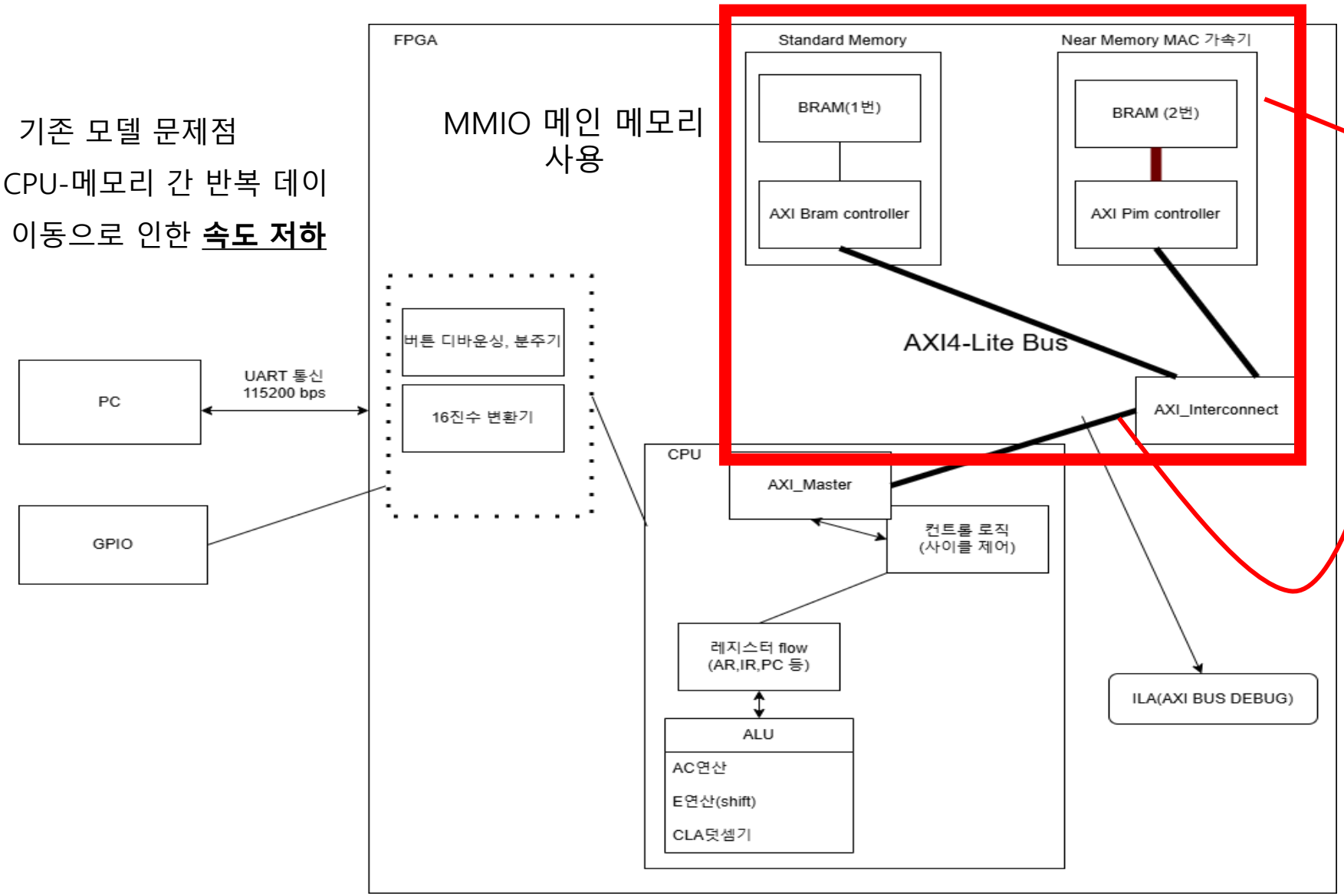
  (2) 데이터값 및 연산모드 입력받기, 합산, 평균, 중앙값 연산하여 시스템 구축

  (3) 커스텀한 CPU와 Xilinx사의 Bram Memory Controller IP와 연결. 버스 프로토콜로는 AXI4-Lite 사용 – 직접 만든 Master와 표준 Slave간의 연결 확인

  (4) PIM Slave 설계 – 단순 연산 시 CPU의 메모리 병목 해결, MAC 연산으로 AI 가속기로의 확장성 내재

1. [구조 개선] : PIM 도입으로 데이터 병목 해소

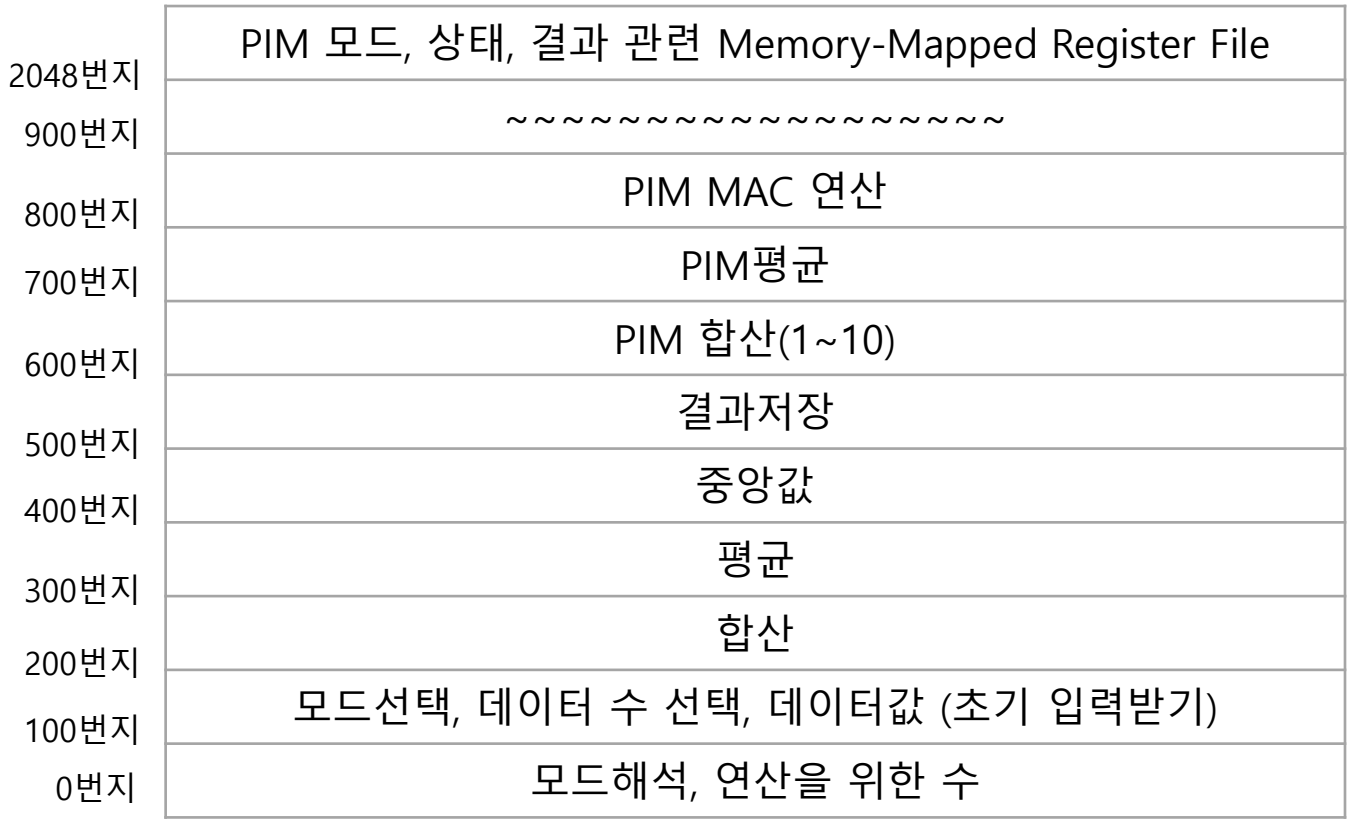
- 기존 모델 문제점
  - CPU-메모리 간 반복 데이터 이동으로 인한 **속도 저하**



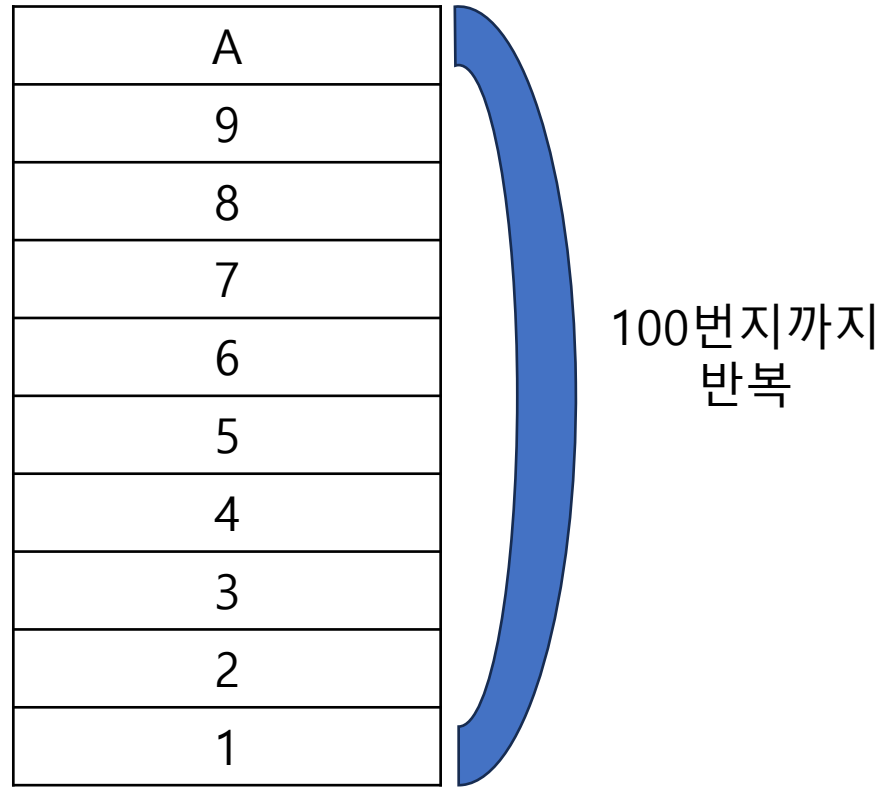
- **PIM 도입**
  - 컨트롤러에 연산기 배치, Wire **이동 거리 감소** 및 병렬 처리 구현

- **AXI4 표준 버스 프로토콜 사용**
  - IP 간의 안정적인 데이터 전송 및 향후 DMA 확장성 확보

2. 메모리맵



메인 메모리

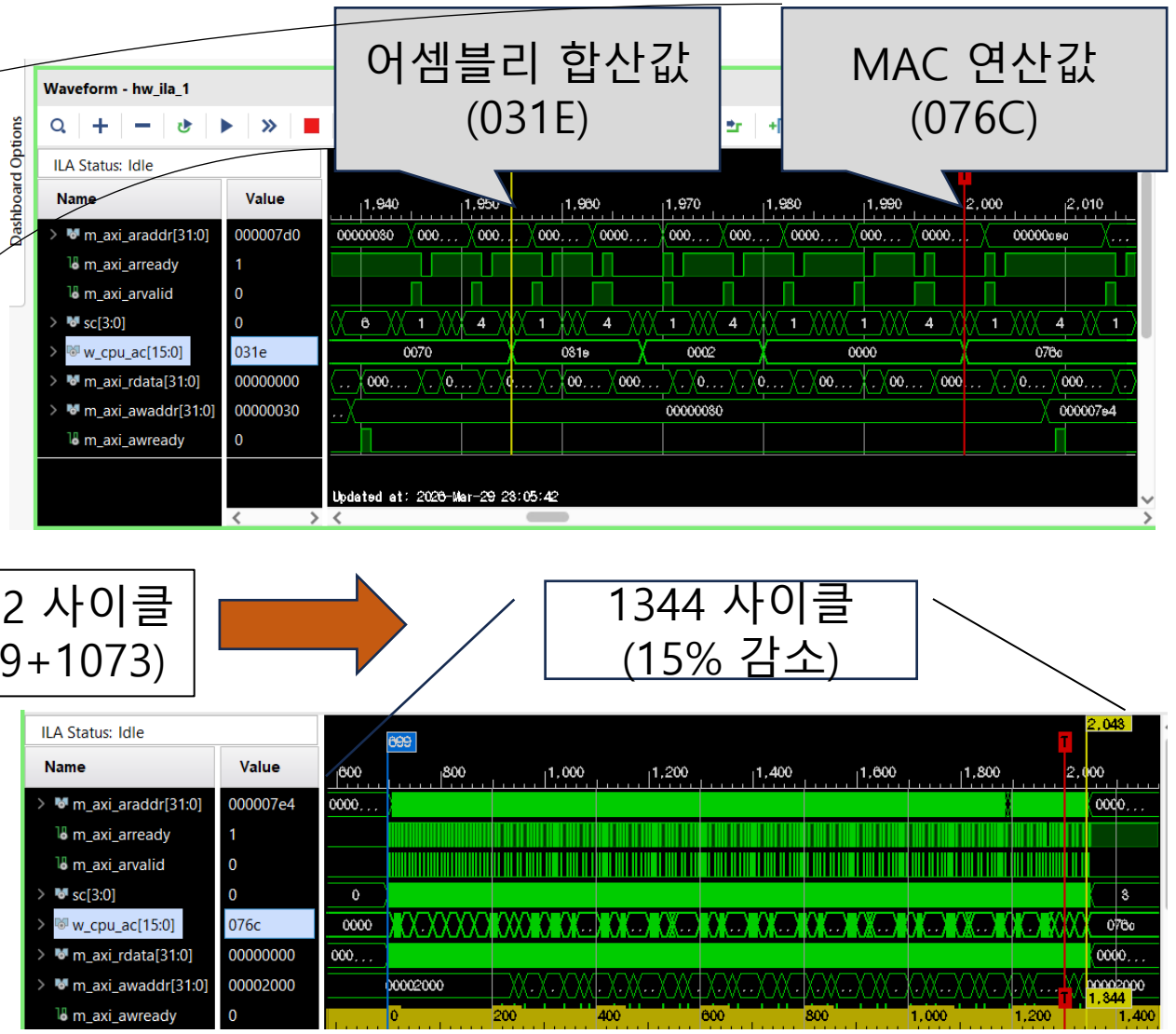


PIM 연동형 온칩 메모리

3. [성능 극대화] : 병렬 처리를 통한 **사이클 15% 단축**

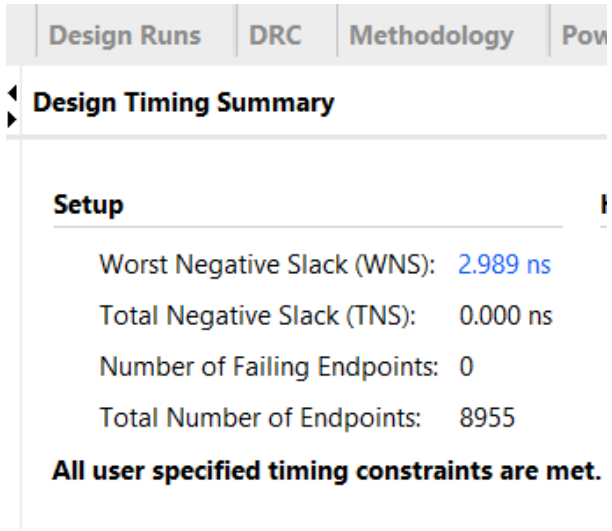


1. FPGA 상의 사이클 비교 – 처리 속도 개선 확인

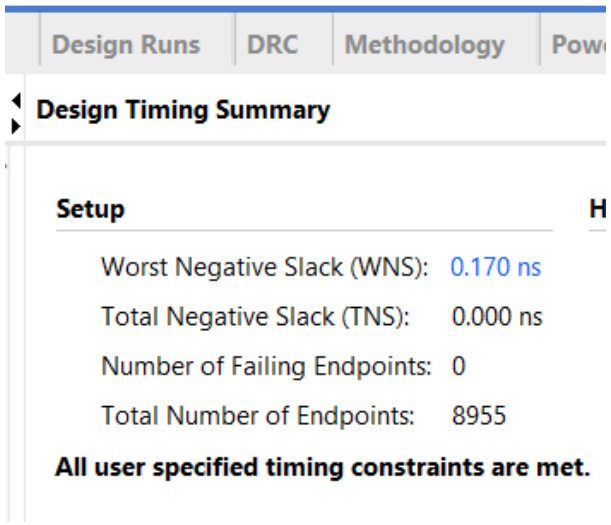


2. ILA 파형 분석 – 병렬처리 사이클 확인

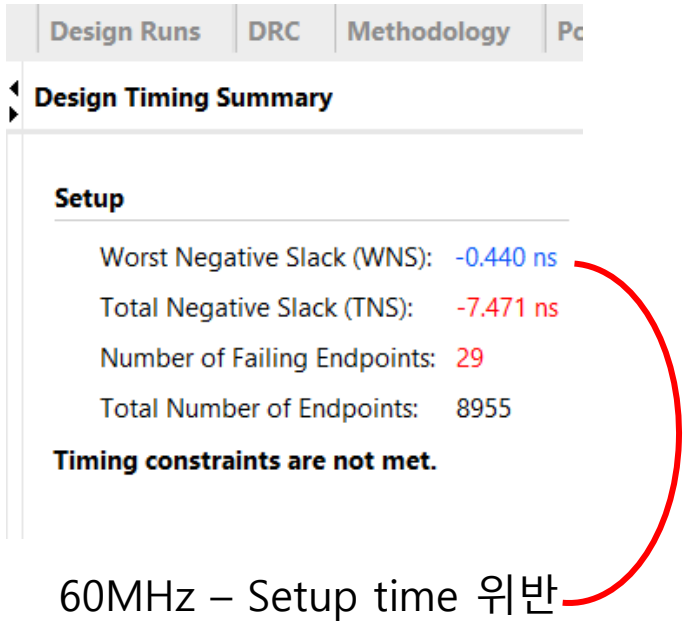
4. [안정성 확보] : 안정된 주파수에서 **성능 달성**



50MHz – 안정된 주파수



58.8MHz – 최대 주파수



60MHz – Setup time 위반

<u>발열 및 타이밍 마진 부족에 의한 주파수 상승 제한 환경에서</u>

<u>PIM 아키텍처로 병렬 처리하여 안정된 주파수 내 고속 처리 달성</u>

PROJECT 02. Ising machine – Max cut

- 수행기간

  2026.07~

- 수행인원

  1명

- 사용기술

  VIVADO(verilog, system verilog), FPGA, ILA, C, Python, UVM

- 수행내용

  (1) 소프트웨어 골든 모델링 및 검증
  (2) 순차 처리 기반 초기 RTL 로직 설계 및 디버깅
  (3) 하드웨어 아키텍처 최적화 (그래프 컬러링 및 병렬화 도입)
  (4) 100MHz 타이밍 클로저 달성 및 FSM 동기화
  (5) UVM 기반 랜덤 seed 회귀 검증 환경 구축

1. 에러율 목표 선정과 실험

하드웨어 | 소프트웨어

Instances	n	d	#	DAv2		DAv3		BURER2002		PAL2004MTS2		MERZ1999GLS	
				# opt	worst acc.	# opt	worst acc.	# opt	worst acc.	# opt	worst acc.	# opt	worst acc.
bqp	100	0.1	10	10	1	10	1	9	0.9973	10	1	10	1
ising	100,200,300	1	30	0	0.9918	6	0.9882	1	0.9869	3	0.9918	4	0.9946
w	100	0.1,0.5,0.9	30	29	0.9956	30	1	30	1	30	1	30	1
t	100–400	0.01025–0.04838	18	9	0.9883	13	0.9881	8	0.9901	2	0.9833	9	0.9819
pm	80,100	0.1,0.99	40	40	1	40	1	40	1	40	1	40	1
gk	20–100	0.0625–1	40	40	1	40	1	35	0.8836	40	1	40	1
g05	60,80,100	0.5	30	30	1	30	1	30	1	30	1	30	1
pw	100	0.1,0.5,0.9	30	30	1	30	1	28	0.9970	30	1	29	0.9975

Table 6: Solver Performance comparison on instances with known optimal solutions from BiqMac Library [71]. Worst accuracy achieved by the solver across all instances in a given class is reported.

#opt : #개수 중 정답을 찾은 횟수
d : 노드 간 연결정도

출처: S. Shaglel 외, "A comprehensive benchmark of an Ising machine on the Max-Cut problem", New Journal of Physics, 2026. (Table 6 발췌)

2. 가중치 메모리 할당

하드웨어

readme.txt | vcu118_rev1.1_12052016.xdc | test.coe

파일 편집 보기

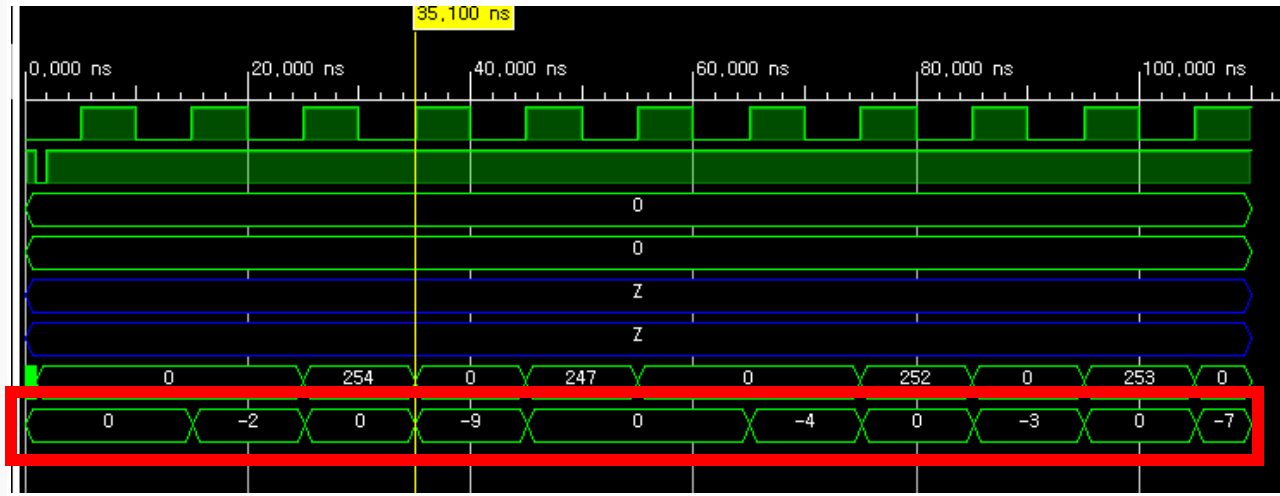
```
memory_initialization_radix=16;
memory_initialization_vector=
00, FE, 00, F7, 00, 00, FC, 00, FD, 00, F9, F7, F7, FD, F9, FB, 00, 00, 00, 00,
FE, 00, F7, 00, FE, FD, FF, FB, FC, FF, F7, FF, 00, FE, FC, FD, F6, F8, 00, 00,
00, F7, 00, F7, 00, F8, F9, 00, FD, FE, FC, FA, FE, FC, FA, 00, F8, FF, 00, 00,
F7, 00, F7, 00, 00, F9, FD, FD, 00, F6, 00, F7, FE, F8, 00, 00, F7, FD, F9, 00,
00, FE, 00, 00, 00, FD, 00, FD, F7, F9, FF, FE, 00, FB, FB, FF, 00, 00, F8, FD,
00, FD, F8, F9, FD, 00, FF, 00, F7, F8, 00, 00, 00, FC, F8, F9, F7, F7, FB, F6,
FC, FF, F9, FD, 00, FF, 00, FD, FA, F7, FD, F8, FE, FF, FF, F7, F6, F8, FE, 00,
00, FB, 00, FD, FD, 00, FD, 00, F9, F6, FE, FE, 00, F8, FC, F7, F8, FA, FC, F8,
FD, FC, FD, 00, F7, F7, FA, F9, 00, FD, FE, 00, 00, 00, 00, F7, 00, FC, F8, 00,
00, FF, FE, F6, F9, F8, F7, F6, FD, 00, F8, FF, 00, FF, 00, FF, 00, FF, FE, FA,
F9, F7, FC, 00, FF, 00, FD, FE, FE, F8, 00, FC, F6, 00, 00, F7, FA, F7, FB, FA,
F7, FF, FA, F7, FE, 00, F8, FE, 00, FF, FC, 00, FC, 00, 00, F6, F9, F7, FE, FB,
F7, 00, FE, FE, 00, 00, FE, 00, 00, 00, F6, FC, 00, FD, FF, FC, F9, FC, 00, 00,
FD, FE, FC, F8, FB, FC, FF, F8, 00, FF, 00, 00, FD, 00, FE, FA, FB, FC, 00, FB,
F9, FC, FA, 00, FB, F8, FF, FC, 00, 00, 00, 00, FF, FE, 00, F7, FC, F6, FB, F6,
FB, FD, 00, 00, FF, F9, F7, F7, F7, FF, F7, F6, FC, FA, F7, 00, 00, FF, F6, FB,
00, F6, F8, F7, 00, F7, F6, F8, 00, 00, FA, F9, F9, FB, FC, 00, 00, FA, FF, FC,
00, F8, FF, FD, 00, F7, F8, FA, FC, FF, F7, F7, FC, FC, F6, FF, FA, 00, F9, 00,
00, 00, 00, F9, F8, FB, FE, FC, F8, FE, FB, FE, 00, 00, FB, F6, FF, F9, 00, F8,
00, 00, 00, 00, FD, F6, 00, F8, 00, FA, FA, FB, 00, FB, F6, FB, FC, 00, F8, 00
;
```

readme.txt | vcu118_rev1.1_12052016

파일 편집 보기

```
memory_initialization_radix=10;
memory_initialization_vector=
0,-2,0,-9,0,0,-4,0,-3,0,-7,-9,-9,-3,-7,-5,0,0,0,0,
-2,0,-9,0,-2,-3,-1,-5,-4,-1,-9,-1,0,-2,-4,-3,-10,-8,0,0,
0,-9,0,-9,0,-8,-7,0,-3,-2,-4,-6,-2,-4,-6,0,-8,-1,0,0,
-9,0,-9,0,0,-7,-3,-3,0,-10,0,-9,-2,-8,0,0,-9,-3,-7,0,
0,-2,0,0,0,-3,0,-3,-9,-7,-1,-2,0,-5,-5,-1,0,0,-8,-3,
0,-3,-8,-7,-3,0,-1,0,-9,-8,0,0,0,-4,-8,-7,-9,-9,-5,-10,
-4,-1,-7,-3,0,-1,0,-3,-6,-9,-3,-8,-2,-1,-1,-9,-10,-8,-2,0,
0,-5,0,-3,-3,0,-3,0,-7,-10,-2,-2,0,-8,-4,-9,-8,-6,-4,-8,
-3,-4,-3,0,-9,-9,-6,-7,0,-3,-2,0,0,0,0,0,-9,0,-4,-8,0,
0,-1,-2,-10,-7,-8,-9,-10,-3,0,-8,-1,0,-1,0,-1,0,-1,-2,-6,
-7,-9,-4,0,-1,0,-3,-2,-2,-8,0,-4,-10,0,0,-9,-6,-9,-5,-6,
-9,-1,-6,-9,-2,0,-8,-2,0,-1,-4,0,-4,0,0,-10,-7,-9,-2,-5,
-9,0,-2,-2,0,0,-2,0,0,0,-10,-4,0,-3,-1,-4,-7,-4,0,0,
-3,-2,-4,-8,-5,-4,-1,-8,0,-1,0,0,-3,0,-2,-6,-5,-4,0,-5,
-7,-4,-6,0,-5,-8,-1,-4,0,0,0,0,-1,-2,0,-9,-4,-10,-5,-10,
-5,-3,0,0,-1,-7,-9,-9,-9,-1,-9,-10,-4,-6,-9,0,0,-1,-10,-5,
0,-10,-8,-9,0,-9,-10,-8,0,0,-6,-7,-7,-5,-4,0,0,-6,-1,-4,
0,-8,-1,-3,0,-9,-8,-6,-4,-1,-9,-9,-4,-4,-10,-1,-6,0,-7,0,
0,0,0,-7,-8,-5,-2,-4,-8,-2,-5,-2,0,0,-5,-10,-1,-7,0,-8,
0,0,0,0,-3,-10,0,-8,0,-6,-6,-5,0,-5,-10,-5,-4,0,-8,0
```

0,-2,0,-9,0,0,-4,0,-3,0,-7,



-10~0까지 임의로 가중치 설정 가독성을 위해 10진수로 변환

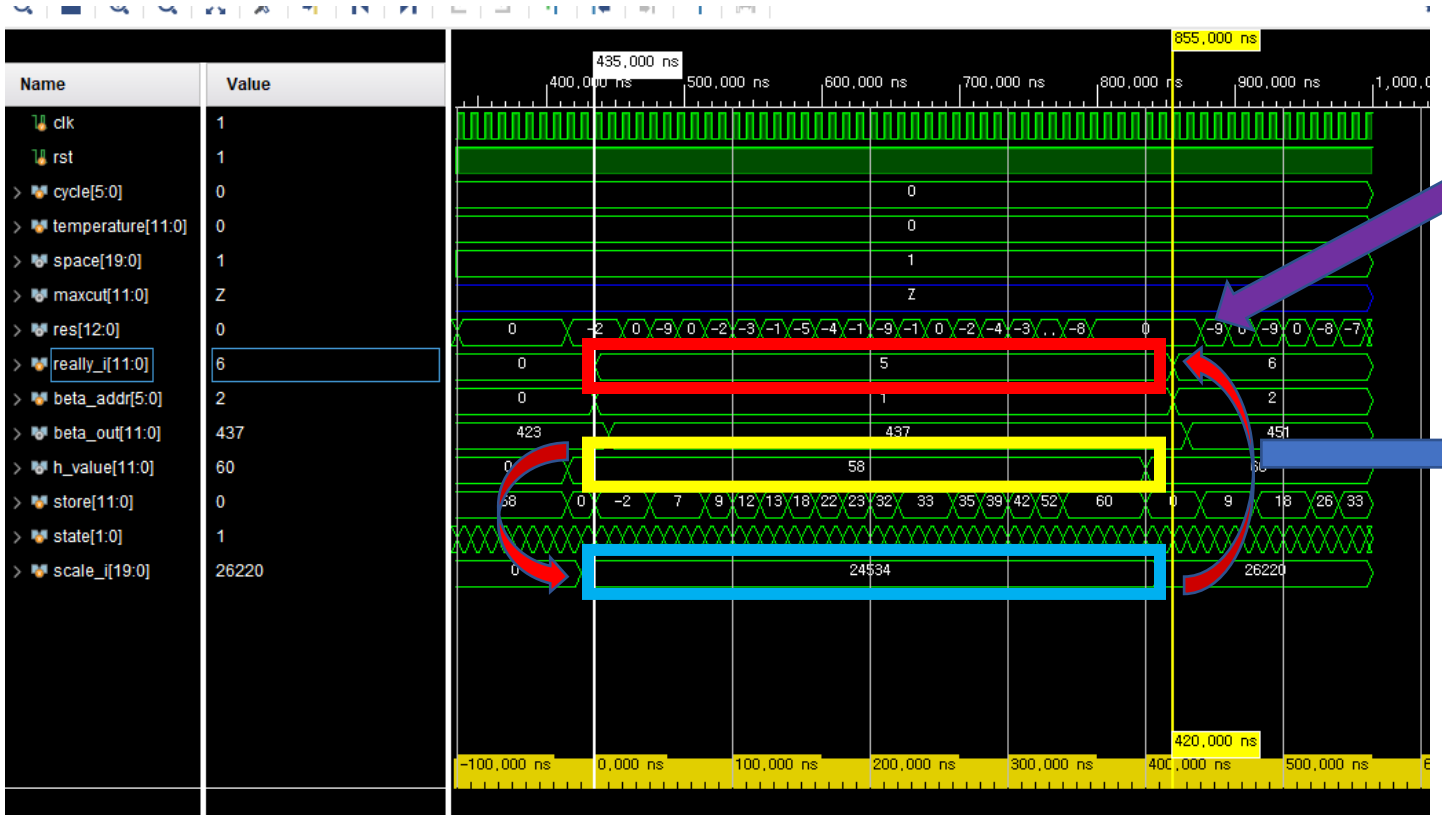
소프트웨어

```
double weight_j[num][num] = { 0.0 };
for (i = 0; i < num; i++) {
    for (j = i + 1; j < num; j++) {
        double rand_weight = -((rand() % 100) / 10.0 + 1.0);
        weight_j[i][j] = rand_weight;
        weight_j[j][i] = rand_weight;   //j_12 = j_21
    }
    weight_j[i][i] = 0.0;
```

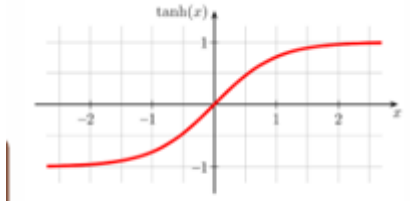
3. Annealing

```
energy energy1(.clk(clk),.rst(rst),.target(),.weight(dout),.space(space),.h_value(h_value),.addr(addr));

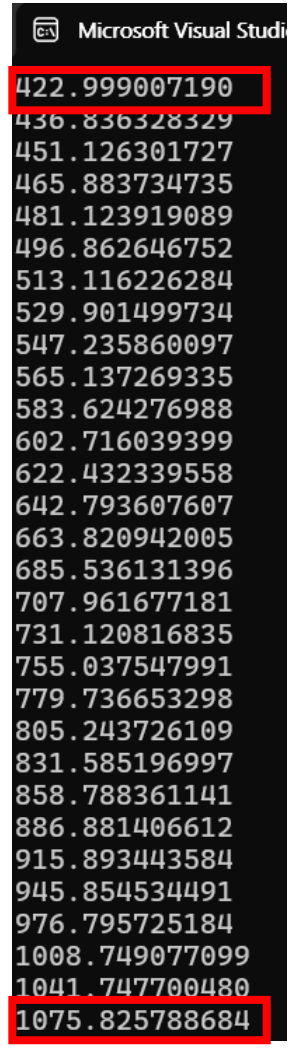
assign scale_i = beta_out*h_value;
assign really_i = scale_i >>> 12;  //>>는 부호유지하기 위해서임. i는 -값 나올 수도 있어서
```



확률적 표현을 위해
베타 처리



tanh함수

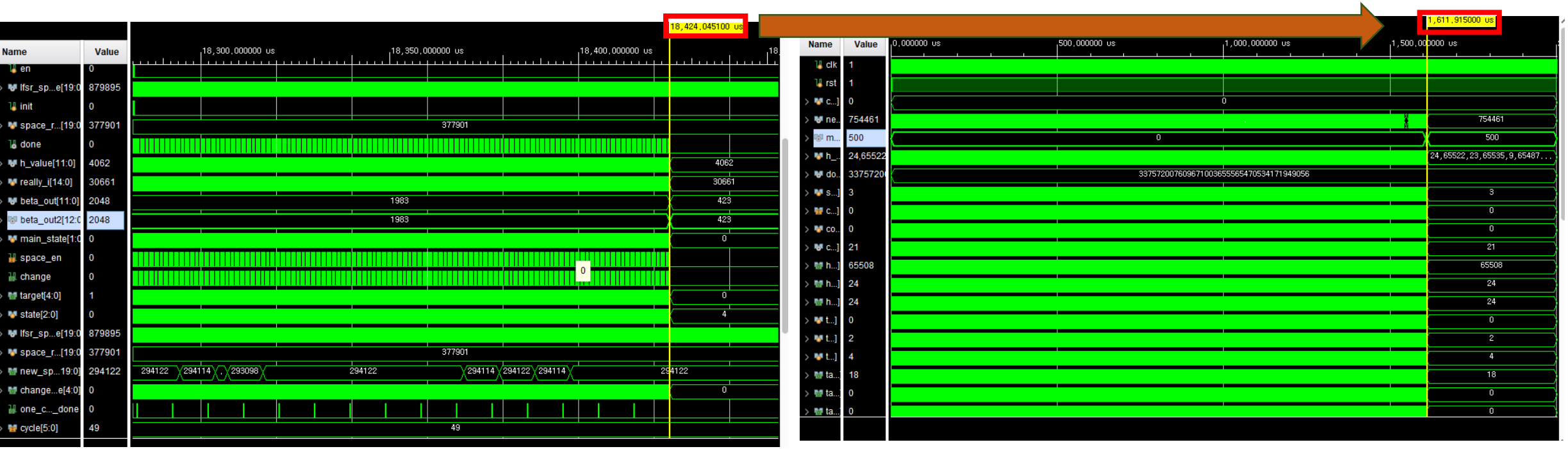


Really_i = 15

4. 타이밍 마진 7.67%, 속도 11.4배 향상

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	0.767 ns	Worst Hold Slack (WHS):	0.055 ns	Worst Pulse Width Slack (WPWS):	4.500 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	1888	Total Number of Endpoints:	1888	Total Number of Endpoints:	989

All user specified timing constraints are met.



5. UVM 기반 랜덤 seed 회귀 검증

Tcl Console | Messages | Log

```
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 941430105000: uvm_test_top.env.sb [LOG] #584 seed=d1081 new_space=0100011111001111010 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 943042155000: uvm_test_top.env.sb [LOG] #585 seed=2f480 new_space=1111101000110000010 maxcut=499
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 944654205000: uvm_test_top.env.sb [LOG] #586 seed=d63a0 new_space=0000010111001111010 maxcut=499
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 946266235000: uvm_test_top.env.sb [LOG] #587 seed=c8605 new_space=0000011111011111010 maxcut=498
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 947878285000: uvm_test_top.env.sb [LOG] #588 seed=03bd9 new_space=0000010111001111010 maxcut=499
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 949490335000: uvm_test_top.env.sb [LOG] #589 seed=894df new_space=0100011111001100010 maxcut=500
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 951102365000: uvm_test_top.env.sb [LOG] #590 seed=57d5b new_space=0100011111001101010 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 952714415000: uvm_test_top.env.sb [LOG] #591 seed=70526 new_space=1111101000100000101 maxcut=499
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 954326465000: uvm_test_top.env.sb [LOG] #592 seed=d3434 new_space=1011100000110001101 maxcut=500
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 955938515000: uvm_test_top.env.sb [LOG] #593 seed=7fe94 new_space=1111101000100000101 maxcut=499
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 957550485000: uvm_test_top.env.sb [LOG] #594 seed=76035 new_space=1011101010110001101 maxcut=442
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 959162515000: uvm_test_top.env.sb [LOG] #595 seed=b613b new_space=0100011111001101010 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 960774565000: uvm_test_top.env.sb [LOG] #596 seed=3f585 new_space=0000010111001111010 maxcut=499
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 962386595000: uvm_test_top.env.sb [LOG] #597 seed=13d3d new_space=0100010111001111010 maxcut=497
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 963998625000: uvm_test_top.env.sb [LOG] #598 seed=31026 new_space=0000010011001111010 maxcut=489
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 965610675000: uvm_test_top.env.sb [LOG] #599 seed=98667 new_space=1011100000110001101 maxcut=500
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 967222705000: uvm_test_top.env.sb [LOG] #600 seed=493ec new_space=0100011111001101010 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 968834735000: uvm_test_top.env.sb [LOG] #601 seed=444e9 new_space=0100011111001101010 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 970446765000: uvm_test_top.env.sb [LOG] #602 seed=89120 new_space=1011100000100010101 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 972058795000: uvm_test_top.env.sb [LOG] #603 seed=16ca5 new_space=0100011111001101010 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 973670845000: uvm_test_top.env.sb [LOG] #604 seed=eef5b new_space=1111101000100000101 maxcut=499
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 975282875000: uvm_test_top.env.sb [LOG] #605 seed=11daa new_space=0100011111001101010 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 976894905000: uvm_test_top.env.sb [LOG] #606 seed=7d713 new_space=0100011111001101010 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 978506935000: uvm_test_top.env.sb [LOG] #607 seed=32bd4 new_space=0100011111001101010 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 980118965000: uvm_test_top.env.sb [LOG] #608 seed=323e5 new_space=0100011111001101010 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 981730995000: uvm_test_top.env.sb [LOG] #609 seed=7c34b new_space=0100011111001101010 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 983343025000: uvm_test_top.env.sb [LOG] #610 seed=0f621 new_space=1011100000100010101 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 984955075000: uvm_test_top.env.sb [LOG] #611 seed=eb813 new_space=0100011111001100010 maxcut=500
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 986567105000: uvm_test_top.env.sb [LOG] #612 seed=12d70 new_space=1011100000100010101 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 988179155000: uvm_test_top.env.sb [LOG] #613 seed=82a0f new_space=0100011111001100010 maxcut=500
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 989791185000: uvm_test_top.env.sb [LOG] #614 seed=3da0e new_space=1011100000100010101 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 991403235000: uvm_test_top.env.sb [LOG] #615 seed=4115c new_space=1011100001001011101 maxcut=484
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 993015265000: uvm_test_top.env.sb [LOG] #616 seed=8f7fb new_space=1011100000100010101 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 994627295000: uvm_test_top.env.sb [LOG] #617 seed=279e4 new_space=0100011111001101010 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 996239325000: uvm_test_top.env.sb [LOG] #618 seed=0931a new_space=1011100000100010101 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 997851355000: uvm_test_top.env.sb [LOG] #619 seed=347ba new_space=1011100000100010101 maxcut=501
UVM_INFO C:/Users/c/Desktop/digital_circuit/ising_machine_color/ising_machine_color.srcs/sim_1/new/verification.sv(185) @ 999463405000: uvm_test_top.env.sb [LOG] #620 seed=a9790 new_space=1111101000100000101 maxcut=499
run: Time (s): cpu = 00:03:38 ; elapsed = 00:15:15 . Memory (MB): peak = 2103.641 ; gain = 0.000
```

지표	수치 (표본 : 620개)
최적해 도달률	41.1%(255회)
평균 오차율	0.625%
중앙값	500 (정답 : 501)
최소/최대 오차율	0% / 11.78%
표준편차	1.55%

PROJECT 03. Flash ADC

- 수행기간

  2026.07~2026.08

- 수행인원

  1명

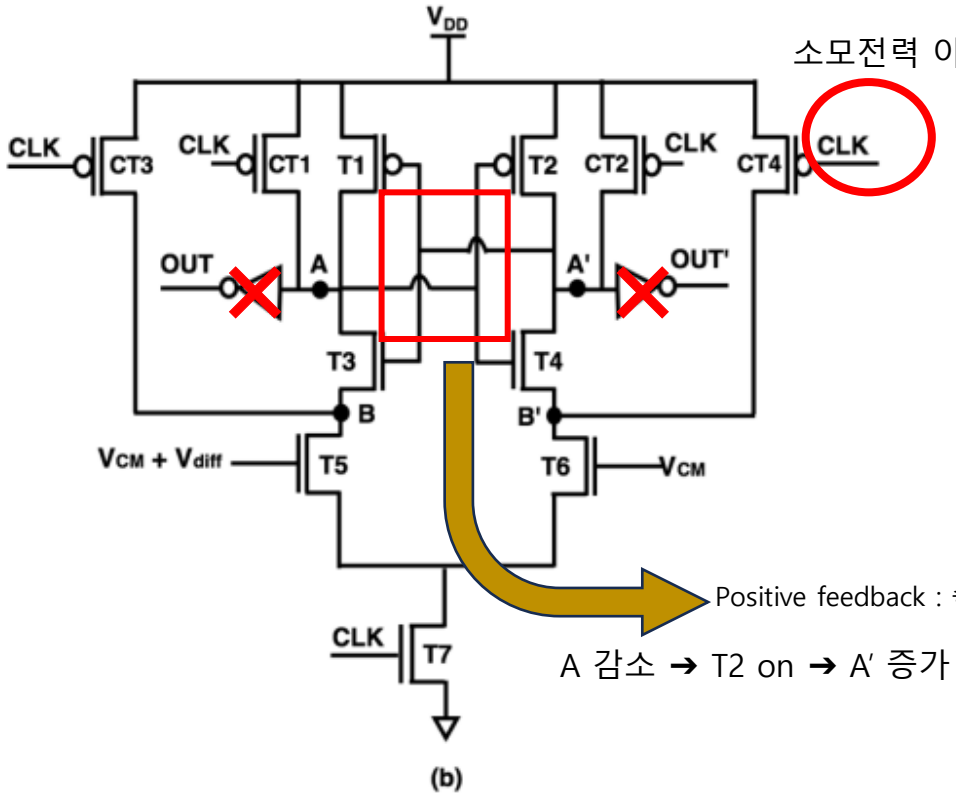
- 사용기술

  Cadence Virtuoso

- 수행내용

  (1) Open-loop 비교기 기반 초기 아날로그 회로 설계 및 검증
  (2) 속도-전력 Trade-off 분석 및 하드웨어 아키텍처 최적화
  (3) 전압 강하 및 플로팅 노드 이슈 디버깅
  (4) 최종 정적·동적 성능 검증 및 타겟 스펙 달성 (INL 0.4mV, ENOB 2.95bit, SNR 19.6dB, SFDR 27dBc 확보)

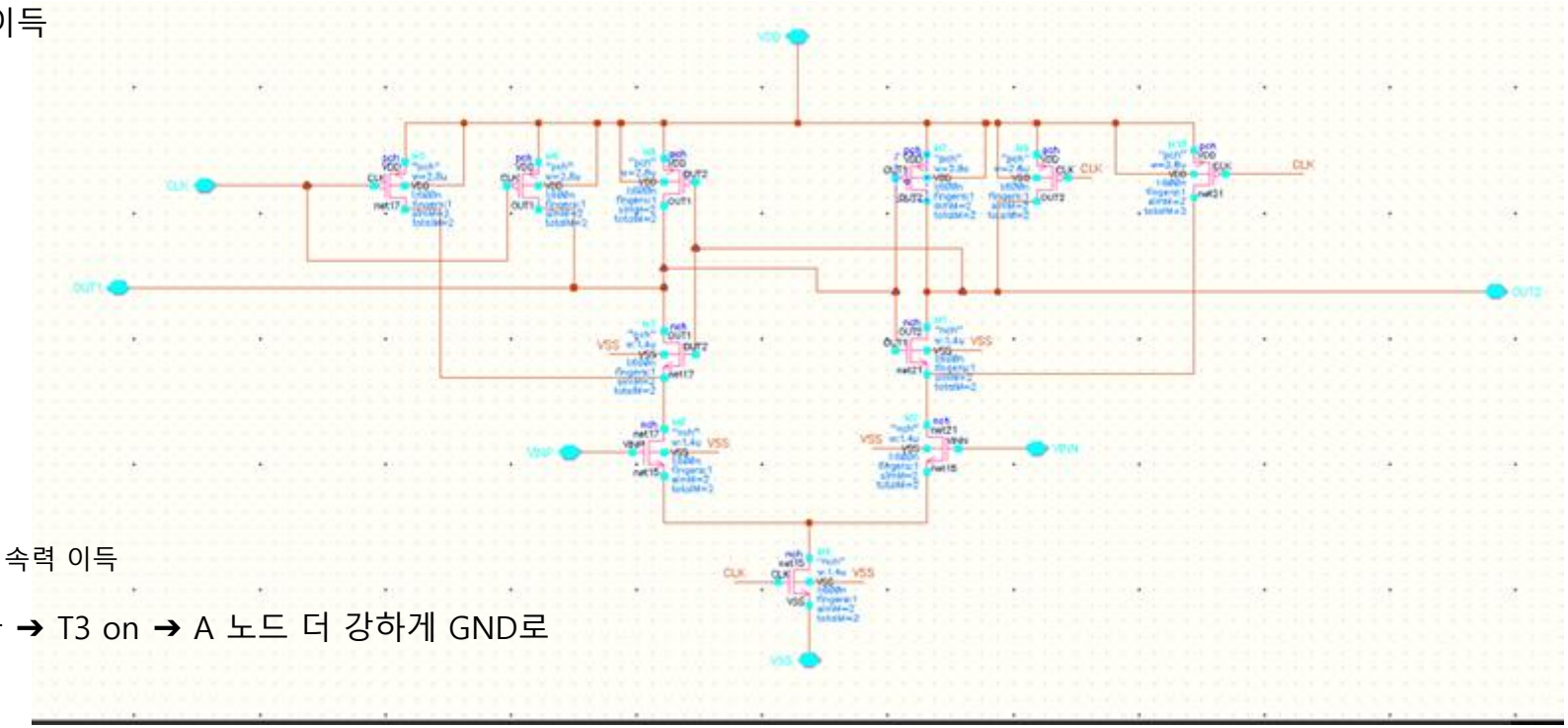
1. StrongARM 도입



소모전력 이득

Positive feedback : 속력 이득

A 감소 ➔ T2 on ➔ A′ 증가 ➔ T3 on ➔ A 노드 더 강하게 GND로



by [4], and (b) conventional design proposed by [5]

출처 : A. Almansouri, A. Alturki, A. Alshehri, T. Al-Attar, and H. Fariborzi, "Improved StrongARM Latch Comparator: Design, Analysis and Performance Evaluation," *Proceedings of the IEEE PRIME 2017*, Giardini Naxos–Taormina, Italy, 2017,

2. Encoder – 이진 트리 확장성

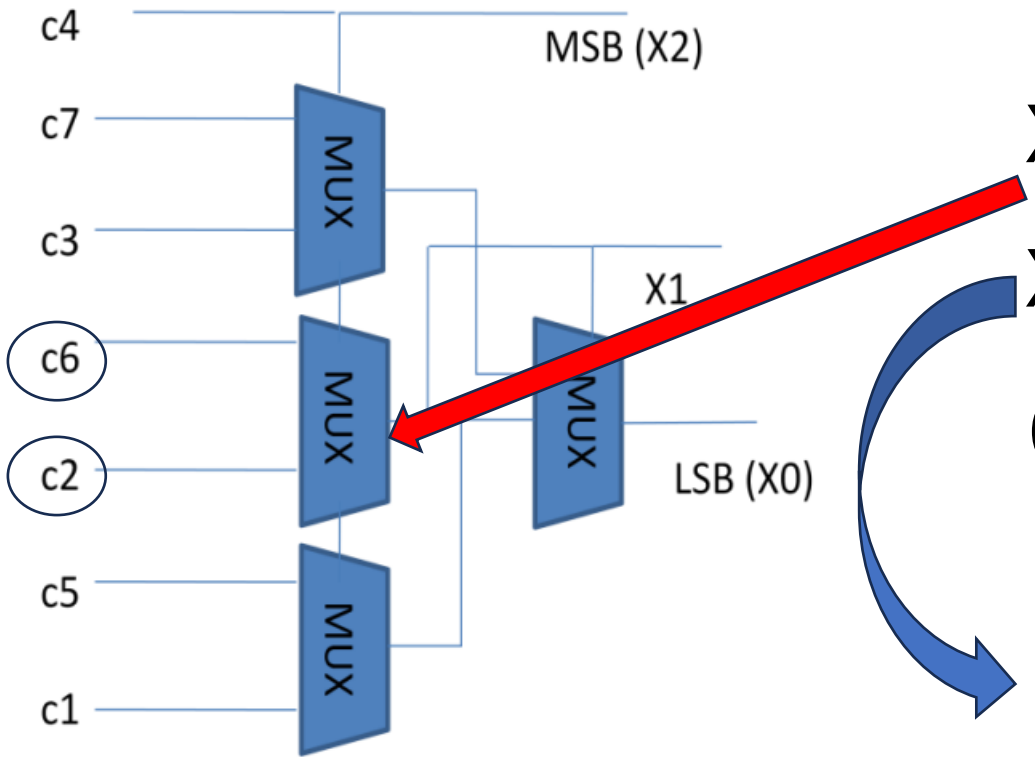


Fig 3: Logic Encoder for 3bit ADC

$X1 = C2\overline{C4} + C6$ (C4가 sel신호)

$X0 = C1\overline{C2} + C3\overline{C4} + C5\overline{C6} + C7$

$(C2, C4, C6$가 sel신호)

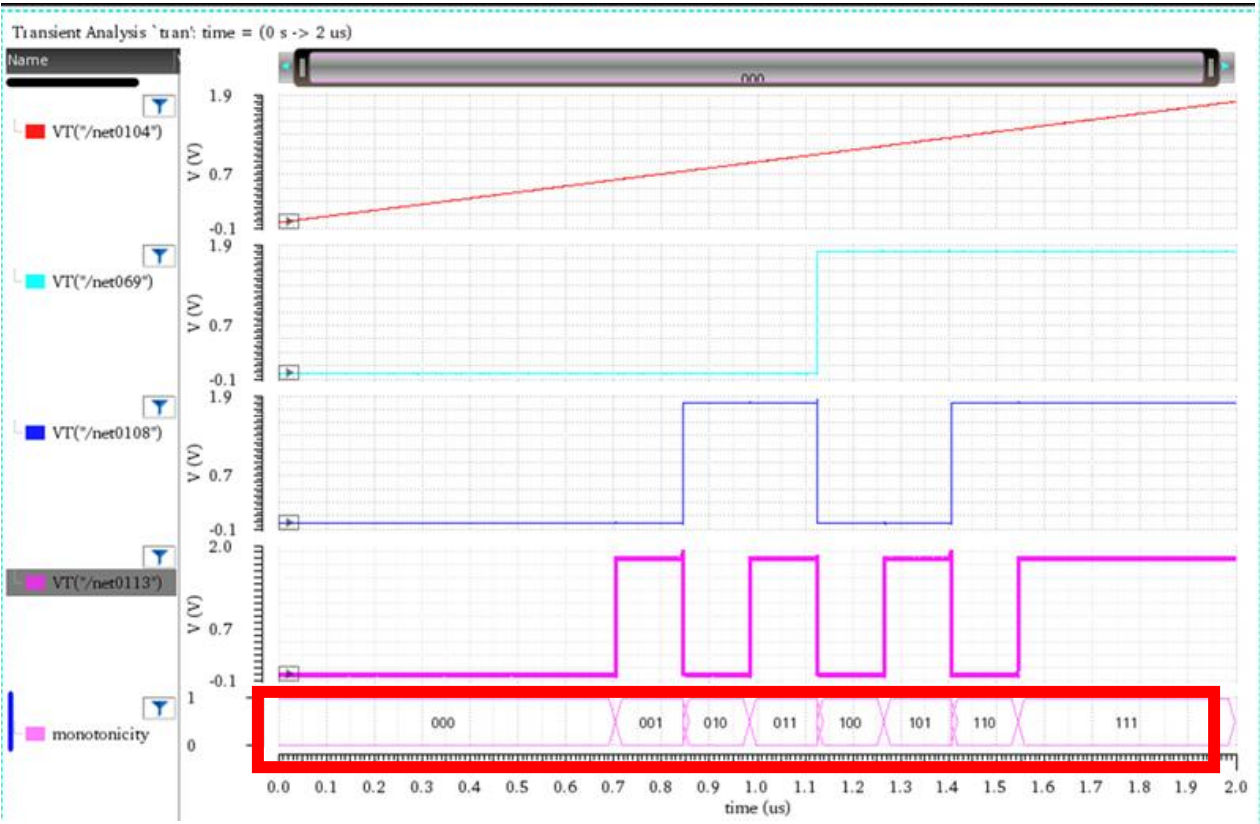
C4=0 : $C1\overline{C2} + C3$ (C2=0 : C1, C2=1 : C3)

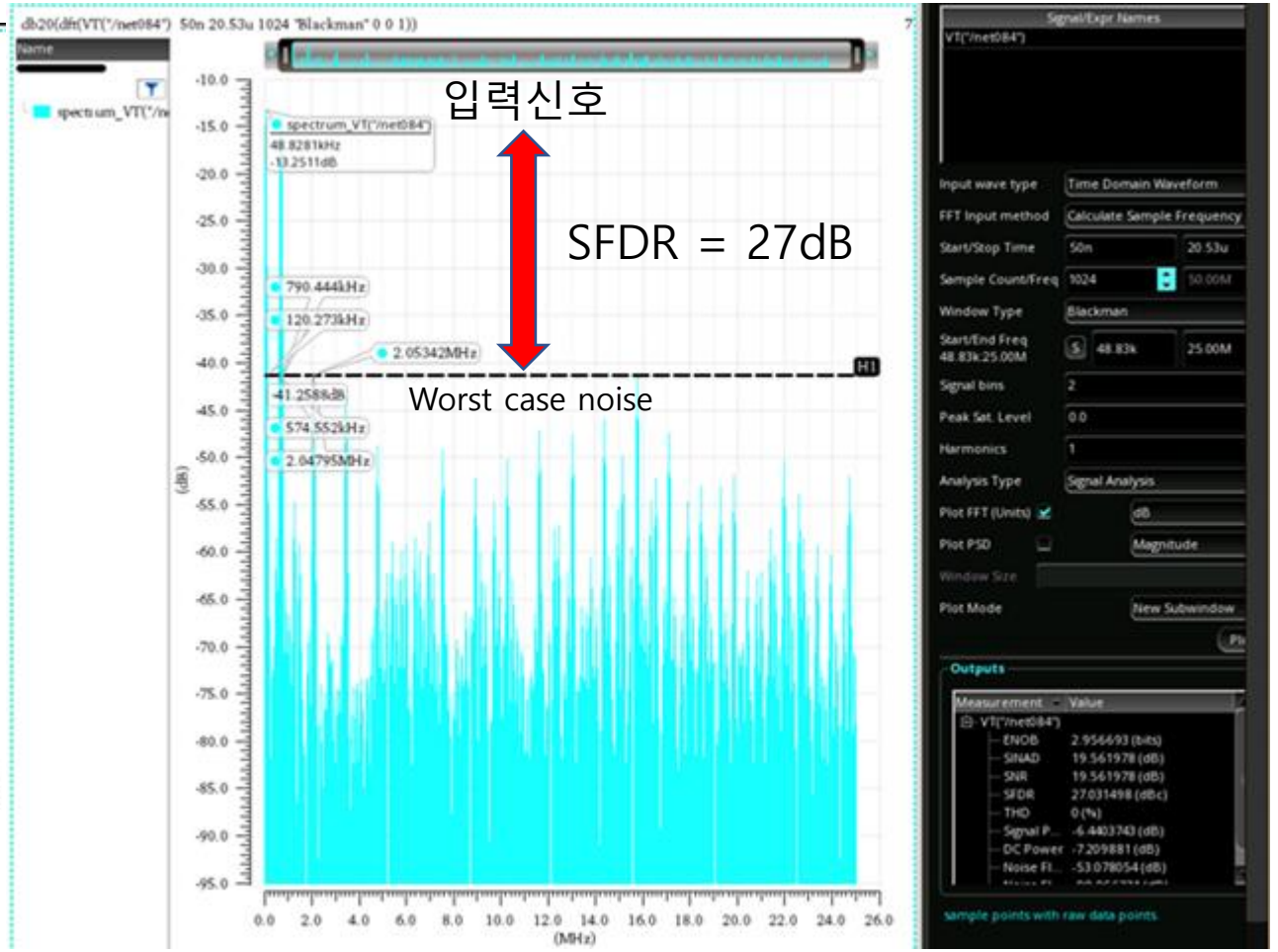
C4=1 : $C5\overline{C6} + C7$ (C6=0 : C5, C6=1 : C7)

출처 : Arunkumar P Chavan, Rekha G, P Narashimaraja, "An Efficient Design of 3bit and 4bit Flash ADC," International Journal of Computer Applications, Vol. 61, No. 11, pp. 32-37, January 2013.

3. 정적, 동적 스펙 검증

ENOB	2.956693bit
SINAD	19.561978dB
SNR	19.561978dB
SFDR	**27.031498dBc**
THD	0%





PROJECT 04. SAR ADC

- 수행기간

  2026.09~2026.10

- 수행인원

  1명

- 사용기술

  Cadence Virtuoso

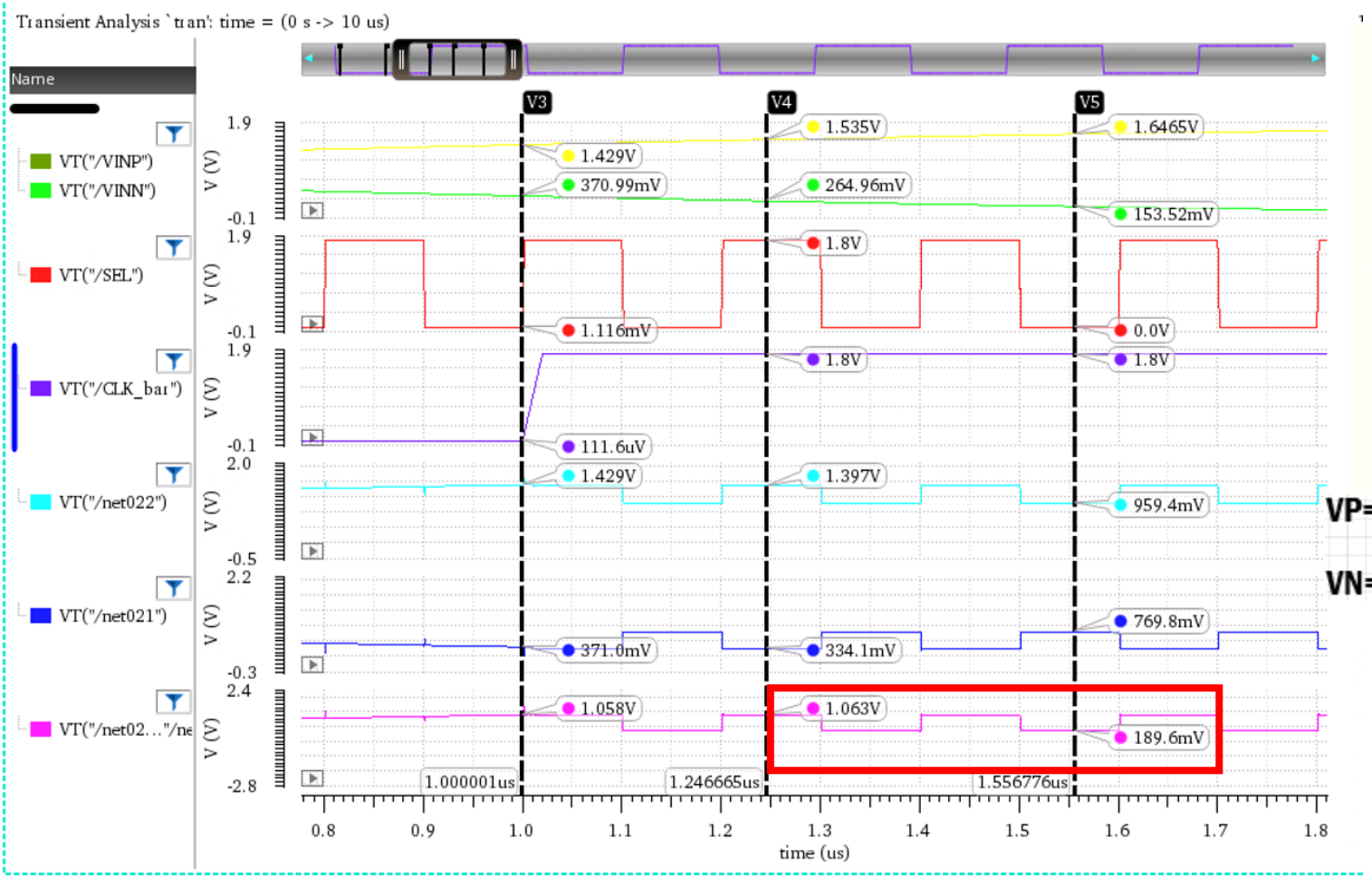
- 수행내용
  (1) 20MHz 샘플링 목표에 맞춰 정착 오차 0.5LSB 기준으로 RC와 유닛 캐패시터 스펙 역산
  (2) 차동 VCM-based CDAC를 2-bit부터 8-bit까지 단계적으로 확장 및 검증
  (3) SAR 로직용 F/F 3종 비교 후 TSPC 개조 및 선정
  (4) Flash ADC 비교기 재활용 후 20MHz 타이밍 디버깅으로 ENOB 5.6bit → 7.75bit 개선

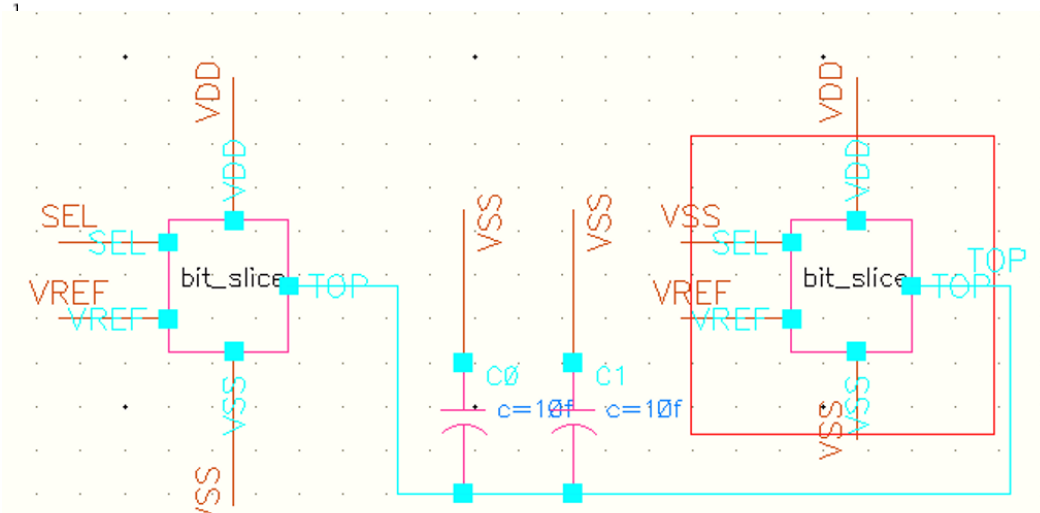
1. 목표 스펙

RC 최대 = 0.5ns
Bandwidth 최소 = 318MHz

- 캡에 충전되는 최종 전압은 목표치와 0.5LSB 이내
- $V_{REF} = 1.8V$
- 1LSB = $\frac{2V_{REF}}{2^n} = \frac{3.6}{256} = 0.01406 = 14.06\text{mV}$
- 최대 error = $\frac{0.5LSB}{256LSB} * 100 = 0.195\%$
- Tau = RC
- t(sampling 시간) = $\frac{0.0625}{20MHz} = 3.125\text{nsec}$
- Error = $e^{-\frac{t}{Tau}}$, ln(error) = $-\frac{t}{Tau} = -\frac{3.125n}{RC}$
- RC = $-\frac{3.125n}{\ln(0.00195)} = \frac{3.125n}{6.24} = 0.5\text{ns}$ (최대) → 이후 활용
- Bandwidth = $\frac{1}{2\pi * Tau} = \frac{1}{2\pi * 0.5n} = 318\text{MHz}$

2. CDAC : 2-bit 설계 후 8-bit 확장

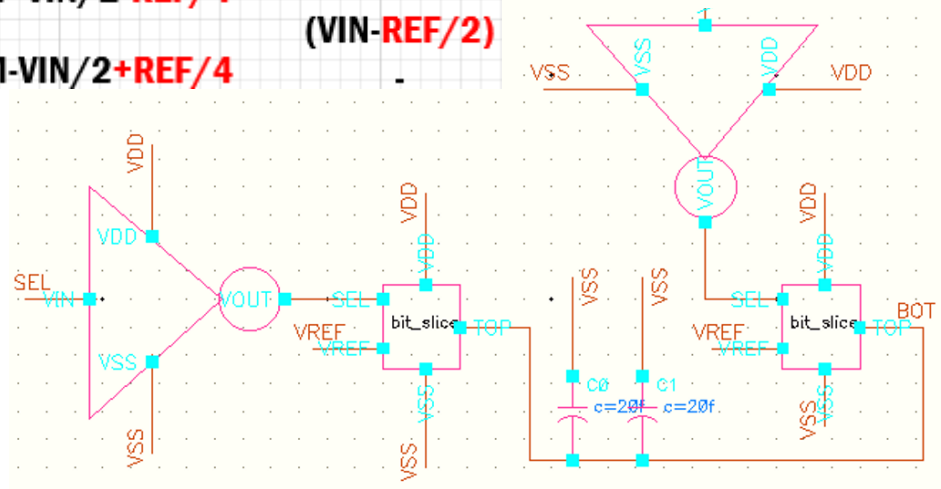




VP=VCM+VIN/2-REF/4

VN=VCM-VIN/2+REF/4

+ (VIN-REF/2) -

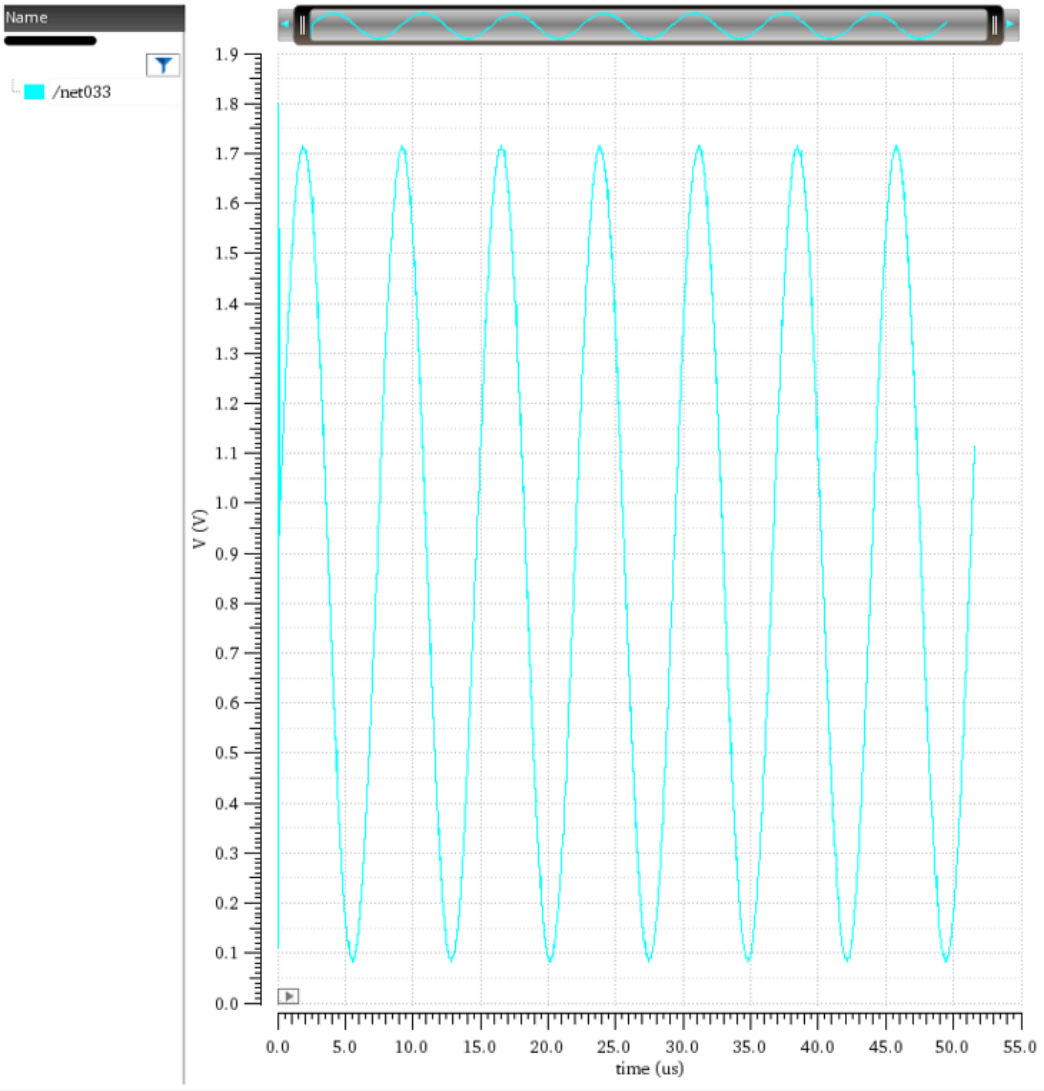


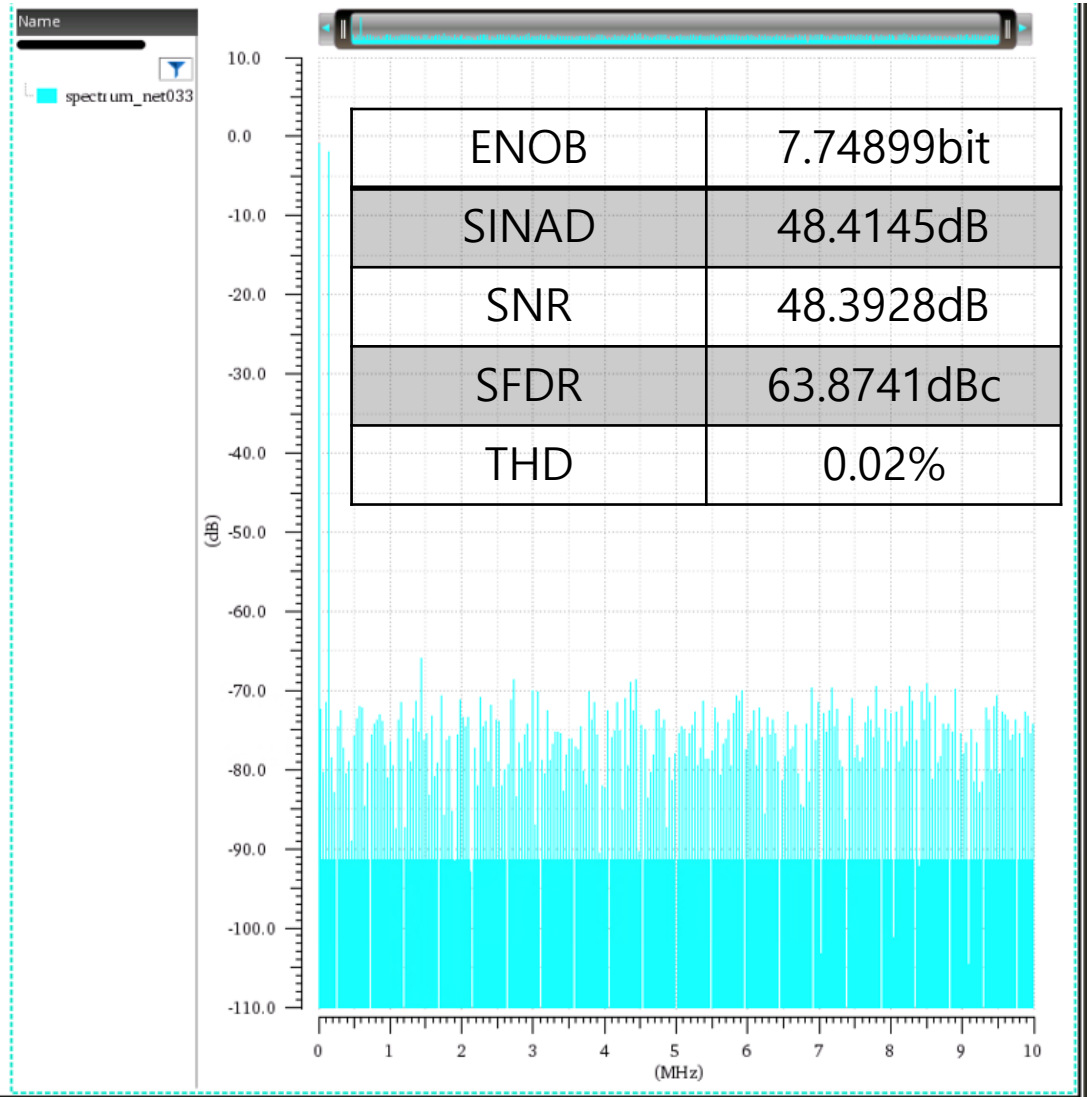
3. SAR 로직용 F/F 3종 비교 후 TSPC 개조 및 선정

에너지 이득

	SR	TG	TSPC	TSPC(개조)
Setup	124.6ps	143.6ps	~~95ps~~	141ps
Hold	0ns이하	0ns이하	~~0ns이하~~	0ns이하
Clk to q	384.2ps	500.2ps	~~366.3ps~~	387.4ps
Setup + clk to q	508.8ps	643.8ps	~~461.3ps~~	528.4ps
1 클럭당 에너지(D고정/D토글)	117.7fJ / 375.9fJ	740.375fJ / 1.019pJ	~~68.314fJ / 279.6fJ~~	82.84fJ / 189.1fJ
Tr 수	26개	20개	~~11개~~	23개

4. 동적 스펙





ENOB	7.74899bit
SINAD	48.4145dB
SNR	48.3928dB
SFDR	63.8741dBc
THD	0.02%



PROJECT 05. FPGA-Based Simple MIPS Processor

- 수행기간

  2026.01

- 수행인원

  1명

- 사용기술

  VIVADO(verilog), FPGA

- 수행내용

  (1) 기본적인 MIPS 명령어 셋을 처리할 수 있는 데이터패스 및 제어 유닛 설계

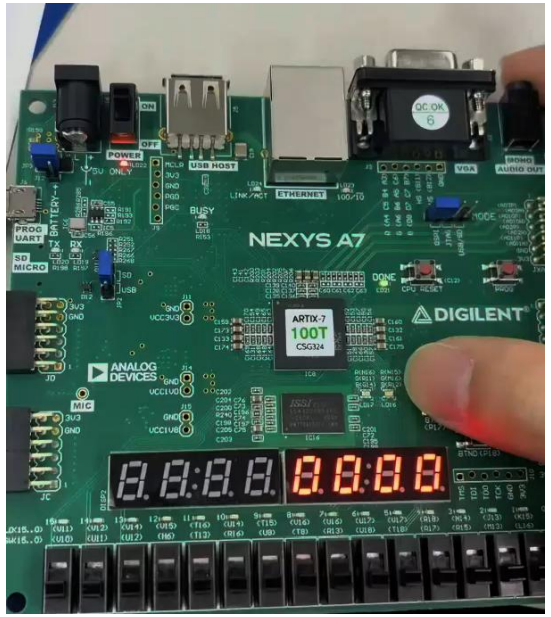
  (2) ALU 내부에 CLA를 직접 설계 및 적용하여, 기존 RCA 대비 덧셈 연산 지연시간 개선

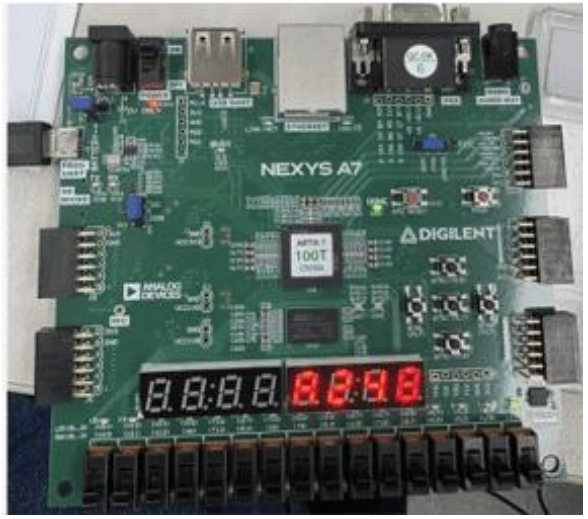
  (3) UART 통신 프로토콜을 구현하여, PC로부터 직접 연산용 데이터를 입력받아 처리하는 I/O 환경 구축

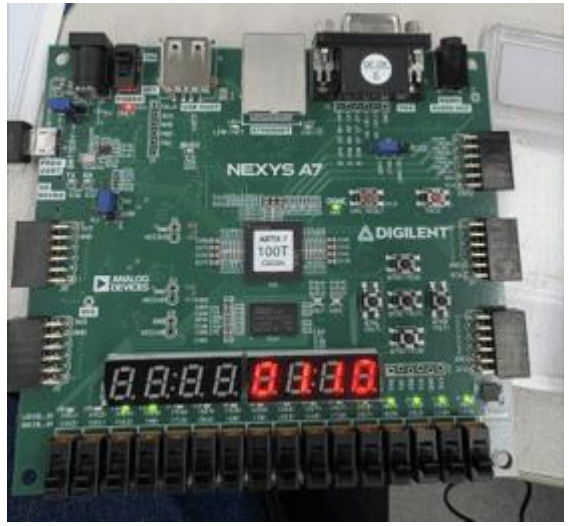
  (4) FSM 기반의 상태 제어 로직을 구현하고, FPGA 보드의 푸시 버튼을 활용한 사이클 단위 디버깅 및 동작 검증 완료

Final Project Milestone1

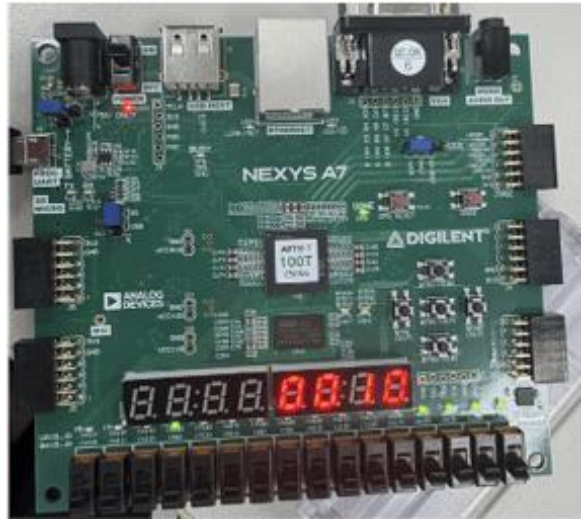
- A248 → B84A
- 시뮬레이션의 결과를 FPGA에서 확인



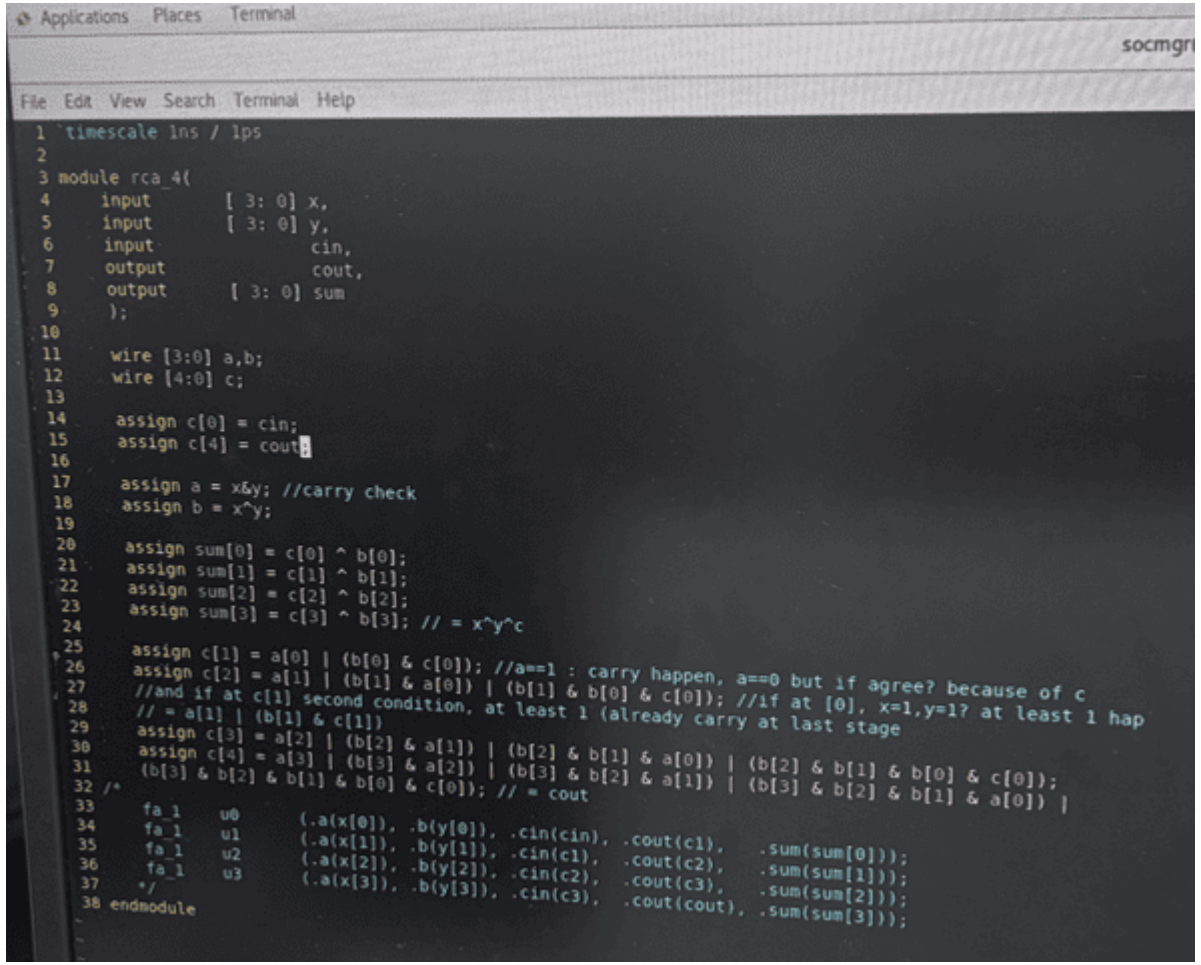








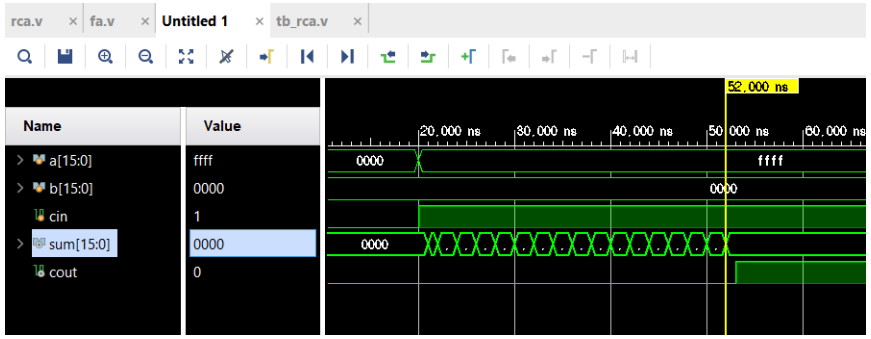
Final Project Milestone2 - CLA : 병렬 carry 계산, 이전 carry직접 사용x

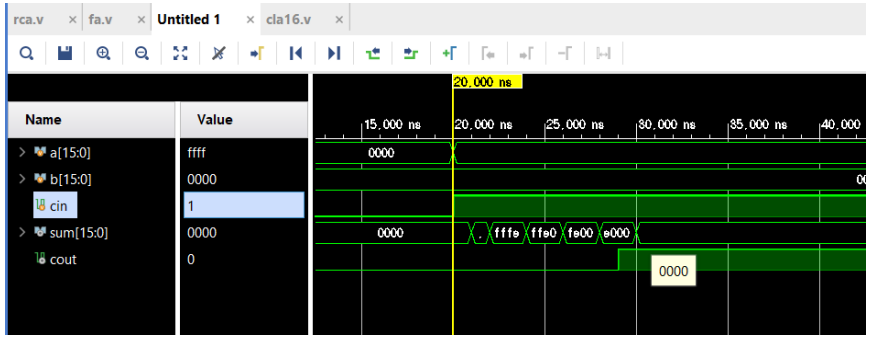


x	y	c	sum	Carry(out)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

- Sum = x,y,c의 xor
- Carry out = xy+c(x^y)
- C[4] = 최종 cout
- 치환사용

RCA – CLA 단일모듈 비교



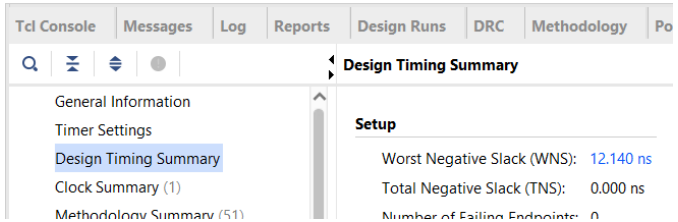


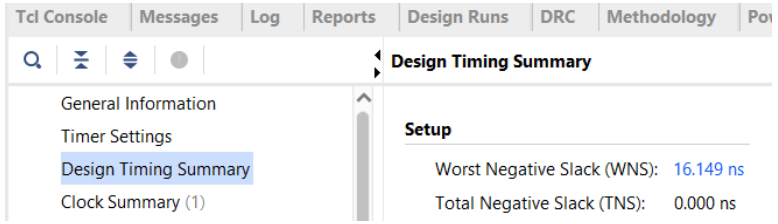
Cout 기준 76% 단축

Resource	Estimation	Available	Utilization...
LUT	16	41000	0.04
IO	50	300	16.67

Resource	Estimation	Available	Utilization...
LUT	29	41000	0.07
FF	50	82000	0.06
IO	52	300	17.33
BUFG	1	32	3.13

LUT 0.03%p 증가





타이밍 마진 33% 추가 확보